

Track 3 — Smart Health & Supply Chain Resilience

AAROGYA

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Github: <https://github.com/mitanshhh/Aarogya>

Video: [Google Drive Link](#)

PROBLEM STATEMENT

Critical Shortages

Unpredictable supply chains lead to undetected out-of-stock vital medicines.

Mismanagement

Inefficient bed tracking causes overcrowding and patient triage delays.

Absenteeism

Severe lack of accountability and staff monitoring in remote rural clinics.

Delayed Analytics

Manual paper reporting drastically slows down district decision-making.

PS: Primary Health Centres (PHCs) across rural India still rely on fragmented records, manual resource management and delayed reporting. This leads to medicine shortages, poor staff accountability, inefficient patient care and slow administrative decisions. Without real-time visibility, district authorities often respond only after problems become critical.

The Healthcare Operations Challenge

Fragmented systems, manual tracking, and reactive decision-making in rural clinics.



Manual Inventory

Paper-based registers lead to hidden stock-outs and delayed restocking for critical medicines. Without digital thresholds, pharmacists cannot predict shortages until the shelves are completely empty, endangering patient lives.



Proxy Attendance

Lack of strict geofenced verification allows absenteeism without immediate detection. Medical Officers manually sign registers for absent peers, meaning remote clinics frequently operate without assigned doctors on-site.



No Central Oversight

District admins lack a unified dashboard to monitor bed occupancy and stock across PHCs. Data sits in isolated physical logbooks, meaning high-level resource reallocation is impossible during sudden regional emergencies.



Reactive Decisions

Facilities act only after supplies run out, lacking ML-driven demand forecasting. Without statistical models analyzing rolling dispense histories, supply chain orders are based entirely on guesswork rather than data.



No AI Insights

Massive amounts of raw data generated without automated executive summaries or risk flagging. The signal is lost in the noise, meaning critical failure points (like sudden disease outbreaks) go entirely unnoticed by administrators.



Isolated Resources

Surplus medicines at one PHC cannot be efficiently requested by a neighboring PHC. Inter-clinic communication requires manual phone calls, leading to vast inefficiencies where medicines expire in one clinic while another faces shortages.

The Aarogya Solution

A full-stack, AI-integrated management platform connecting the entire healthcare hierarchy.



1. Real-Time Visibility

What: Centralized district command view.

How: Live KPIs (Bed occupancy, medicine alerts) aggregated via `/api/v1/district/overview` with 15s caching.

Why: Enables proactive administration.



2. AI Risk Analysis

What: Automated risk identification.

How: Gemini 2.5 Flash processes KPIs to structured JSON. *Fallback:* Rule-based handler if LLM is down.

Why: Turns raw data into executive insights.



3. Automated Audits

What: Monthly AI analysis to PDF.

How: ReportLab generates structured PDFs from Gemini insights containing risks & recommendations.

Why: Standardizes health reporting.



4. QR + GPS Attendance

What: Anti-proxy staff check-ins.

How: 100m geofence. 5-min QR expiry. Random re-verification notifications within a 2-minute window.

Why: Ensures true physical presence.



5. Smart Inventory

What: Automated stock balancing.

How: Predicts stock-outs. Triggers `ResourceRequest` states: PENDING → SHIPPED → COMPLETED.

Why: Prevents medicine shortages.



6. Unique Patient ID

What: Centralized patient record.

How: Links demographics, Bed Allocation, Medicine logs, and an action Audit Timeline via `PT-XXXXXX`.

Why: Unified health history.

Operations Under the Hood

Evidence-based feature modules extracted directly from the codebase.

Inventory Management

Tracks every medical item against its `min_threshold` limits to proactively identify shortages. Issues automated low-stock warnings, feeds into the ML pipeline, and enables seamless bulk CSV uploads (with strict injection prevention) for rapid inventory synchronization.

Routes: `/api/v1/inventory`

Tables: `inventory_items`, `inventory_logs`

PHARMACIST, MO, DIST_ADMIN

Attendance System

Employs the Haversine formula for physical geofence validation alongside an `APScheduler` background process for random anti-proxy checks. Supports multi-entry workflows and instant dashboard refresh every 15 seconds.

Routes: `/api/v1/attendance`

Tables: `daily_qr_sessions`, `attendance_records`

DOCTOR, RECEPTIONIST, LAB_TECH

Bed Allocation

Strict state machine controlling Available/Occupied transitions in real-time. Automatically updates the district command view and is a core 30% weight factor in the deterministic PHC Health Score algorithm.

Routes: `/api/v1/beds`

Tables: `beds`, `patients`

RECEPTIONIST, MO

Patient Audit Timeline

A chronological immutable audit log tracking all VIEW, EDIT, and CREATE events linked to a unique `PT-XXXXXX` patient ID. Fully integrated with ABHA standards and accessible via a unified profile drawer.

Routes: `/api/v1/patients`

Tables: `patients`, `patient_audit_logs`

RECEPTIONIST, DOCTOR, MO

Resource Transfers

A highly controlled cross-PHC state machine managing medicine shortages. Ensures that donor PHC candidates strictly reside within the same hierarchical region to prevent unauthorized cross-border transfers.

Routes: `/api/v1/district/resource-request`

Tables: `resource_requests`

DISTRICT_ADMIN

Automated Reporting

Executes complex background aggregations to feed the Gemini API, returning structural JSON that is piped directly into `ReportLab` to generate executive-ready PDF audit logs containing risks and actionable recommendations.

Routes: `/api/v1/analytics/reports`

Tables: `reports`

MEDICAL_OFFICER, DIST_ADMIN

16 Frontend Pages

~45 API Endpoints

15 DB Tables

9 RBAC Roles

Intelligence & Access Controls

Layered intelligence backed by strict dual-layer Role-Based Access Control.

Forecasting ML (Ridge Regression)

Features: rolling 7d/30d dispenses, days since restock, footfall proxy.
Forecast Horizon: 7 days. Cadence: Weekly via APScheduler.

Gemini 2.5 Flash Insights

Analyzes live operational KPIs, outputting structured JSON with risks and executive summaries. (No custom ML fallback).

Access Matrix (Tenant Scoped)

*Enforced via `require_role` and `permissions.ts`

Module	Dist Admin	Med Officer	Doctor
Dashboard	✓	✗	✗
Inventory	✓	✓	✗
Attendance	✓	✓	✓

Technologies Used

A modern, scalable stack focused on performance and reliability.

FRONTEND

- **React 19.2 & Next.js 16.2**
- TailwindCSS 4
- Shadcn UI & Recharts
- html5-qrcode

BACKEND API

- **Python 3.11 & FastAPI**
- SQLAlchemy 2.0 (ORM)
- Alembic Migrations
- PyJWT + bcrypt Auth
- ReportLab (PDFs)

DATA & ML

- **PostgreSQL on Neon**
- scikit-learn (Ridge Regression)
- Pandas & NumPy

AI & LLMS

- **Gemini 2.5 Flash API**
- Groq (Llama-3.3-70b)
- Regex Fallback Intents

HOSTING & DELIVERY

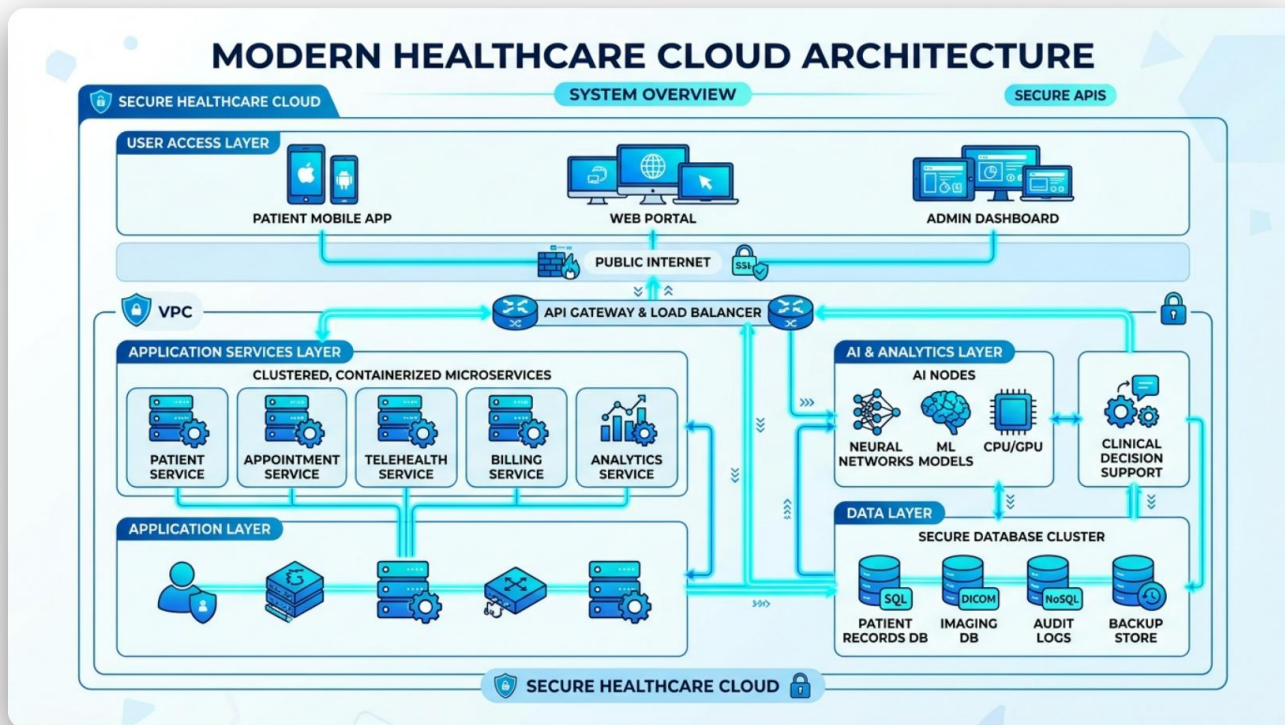
- **Vercel** (Frontend)
- **Render** (Backend API)
- **Neon** (Database)

TOOLING & COMMS

- Google Antigravity AI IDE
- Git & GitHub
- Gmail SMTP

Detailed System Architecture

A full-stack, AI-integrated microservice architecture designed for absolute data isolation and high availability.



Core Operational Workflows

Secured execution paths verified in the codebase.

1. QR + GPS ATTENDANCE SEQUENCE

MO generates QR → Doctor Scans → Haversine < 100m → PRESENT Status → APScheduler (1-4h later) → Random Notif. → 2-min window

Mechanism: The Medical Officer initiates a daily session generating a cryptographically secure, time-sensitive QR code. When staff scan the code, the backend invokes the Haversine formula to validate GPS coordinates against the health centre's fixed location (100m radius). To eliminate proxy attendance, the `APScheduler` cron job triggers randomized re-verifications where staff must respond within a strict 2-minute window or their status defaults to ABSENT.

2. CROSS-DISTRICT MEDICINE TRANSFER

PHC A (Low Stock) → Raises Request → District Admin → Selects PHC B → Status: PENDING_DONOR → PHC B Ships → Status: COMPLETED

Mechanism: Instead of waiting for centralized supply chains, PHCs experiencing acute stock-outs can raise a `ResourceRequest`. The District Admin reviews the request and the AI redistribution engine filters the nearest facilities with surplus stock within the same district boundary. The selected donor PHC receives actionable notifications to approve and ship the required inventory.

3. AUTHENTICATION & RBAC LAYER

Login → bcrypt verify → HttpOnly JWT → require_role() Dependency → Data Scoped by hospital_id

AI & ML Pipelines

Structured integration of deterministic models and LLMs.

1. Executive Analytics Generation

- **Input:** Deterministic DB aggregation (Occupancy, Footfall).
- **Model:** Gemini 2.5 Flash via `google-genai`.
- **Output:** Strictly formatted JSON (Risks, Summary).
- **Safeguard:** Hardcoded JSON fallback if API is unavailable.

2. Demand Forecasting

- **Input:** Rolling means, days since restock, seasonality.
- **Model:** Ridge Regression (`scikit-learn`).
- **Output:** 7-day predicted consumption.
- **Action:** Calculates estimated stock-out date in cache.

3. Natural Language Chatbot

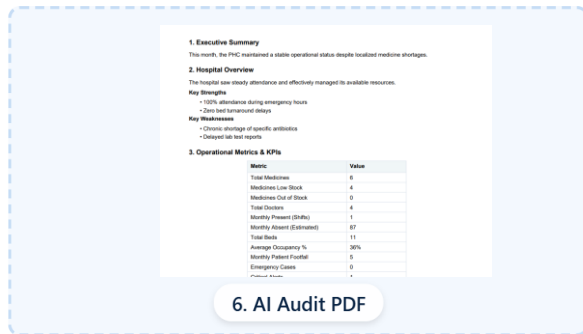
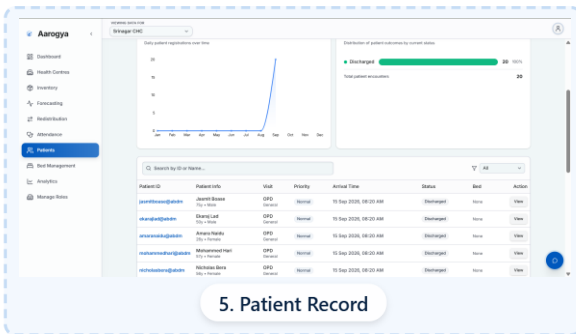
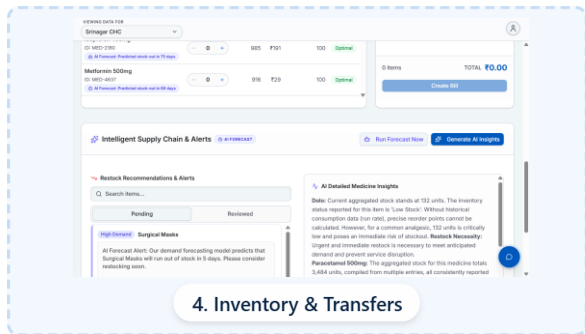
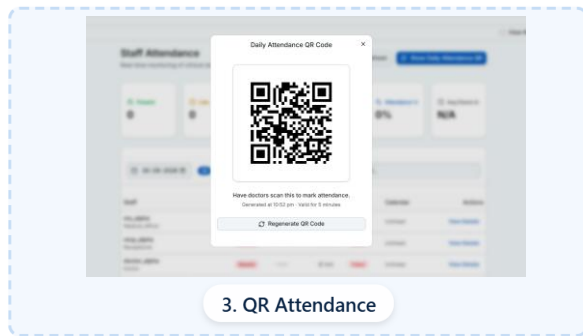
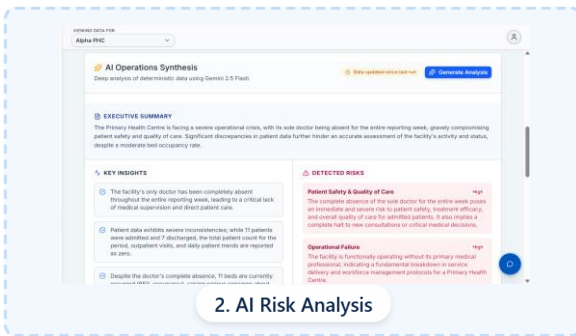
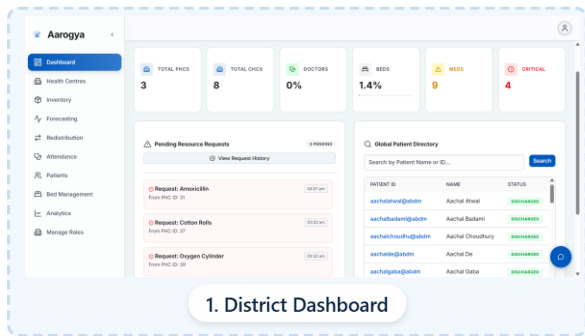
- **Input:** User text queries.
- **Processor:** Groq (`llama-3.3-70b`) or regex rules.
- **Safeguard:** Strict read-only SQL generation blocklist.

4. PHC Health Score Algorithm

- **Input:** Live DB rows.
- **Formula:** 30% Beds + 30% Inventory + 25% Attendance.
- **Action:** Real-time sorting metric for District dashboard.

Product in Action

Live on Vercel · Render · Neon



Security, Scalability & Performance

Enterprise-grade architecture applied to rural healthcare needs.



Security Architecture



- **Authentication:** Passwords securely hashed with `bcrypt`, using `HttpOnly` JWT tokens.
- **Tenancy Isolation:** Absolute `hospital_id` scoping injected dynamically into every SQL query.
- **Physical Verification:** Haversine GPS geofence mathematically enforces 100m proximity bounds.
- **AI Safety Guards:** Regex intent parsing coupled with a strict SQL blocklist guarantees read-only LLM execution.



Scalability Framework

- **Stateless API Design:** The FastAPI backend is entirely stateless, allowing infinite horizontal scaling on Render with automated load balancing.
- **Federated Architecture:** The BRICS FedAvg node aggregator deliberately prevents raw data transmission, completely bypassing central bandwidth bottlenecks and protecting data sovereignty.
- **Async Processing:** Complex operations like the weekly Ridge Regression ML demand forecasting are offloaded entirely to background workers via `APScheduler` cron jobs, ensuring zero impact on the critical path response times of clinical APIs.



Performance Optimization

- **Database Optimization:** Mission-critical Postgres columns (e.g., `patient_code`, `status`, `hospital_id`) are heavily indexed using SQLAlchemy constraints, enabling microsecond querying on Neon.
- **Caching Strategy:** The heavy district overview dashboard endpoint is temporarily cached for 15s to absorb multi-admin spike traffic. ML inferences are pre-computed weekly and read instantly.
- **Data Volume Control:** CSV Bulk upload routes enforce a strict 1000-row limit with immediate memory-safe sanitization, preventing out-of-memory crashes on cheap PHC hardware.

Impact & Roadmap

From local PHC operational efficiency to BRICS-wide data sovereignty.

Fit for Today's Realities

The deployed Aarogya architecture gracefully handles extreme real-world constraints inherent to rural deployments. The mathematically verified 100m Haversine geofence completely eliminates proxy attendance with zero specialized hardware. Highly-tuned Role-Based Access Control (RBAC) ensures that a PHC Pharmacist views an optimized inventory-only UI, while District Admins command the entire aggregated regional view instantaneously.

National Roadmap

Planned

- **Ensemble AI Upgrades:** Transitioning from linear Ridge Regression to complex XGBoost & LightGBM trees.
- **Database Scale-Out:** Postgres read replicas and geospatial sharding.
- **Field Adaptability:** Progressive Web App (PWA) offline-first syncing capabilities for deep rural ASHAs.

BRICS Federation Aggregator

Built



A standalone microservice executing mathematically-proven **Federated Averaging (FedAvg)** ensures highly sensitive patient metrics never cross international borders, strictly preserving BRICS data sovereignty.

1. **Local Node:** Ridge model trains directly on siloed PHC data.
2. **Transmission:** `/push-model` endpoint transmits only anonymous `coef` and `intercept` weight vectors.
3. **Aggregation:** Central Node computes the mathematical global mean.
4. **Synchronization:** `/pull-model` endpoint broadcasts the fortified Global Model back.